SPINNI: Sagnac ParviInterferometer the Next New Innovation

Annie Chiu, Malvika Dias, Mara Portillo, Amy Steinhardt

Introduction

Research Question

- “To what extent can we miniaturize the Sagnac Interferometer system? (portable?)”

Literature Review

- “Gyroscope measurements of the precession and nutation of Earth’s axis”
  - Sagnac interferometer measures **planetary precession and nutation**
  - Sagnac interferometer in the article has dimensions of 4x4 meters

The Sagnac Interferometer & Effect

- Sagnac interferometer
  - Splits beam into opposite directions
  - Across same circular path
  - Spinning surface
- Sagnac effect
  - Laser detector anchored to a spinning surface
  - Spin causes split beams reach detector at different times → **lengths of paths differ**
  - Resulting interference **(phase shift)** → Sagnac effect

Methods

Astro

- Research journal
- Intended to measure planetary rotational quantities

Quantum

- Seismic isolation table
- HeNe laser
- Wave plates
  - Quarter plates
  - Half plates
- Beam splitter
- Detector

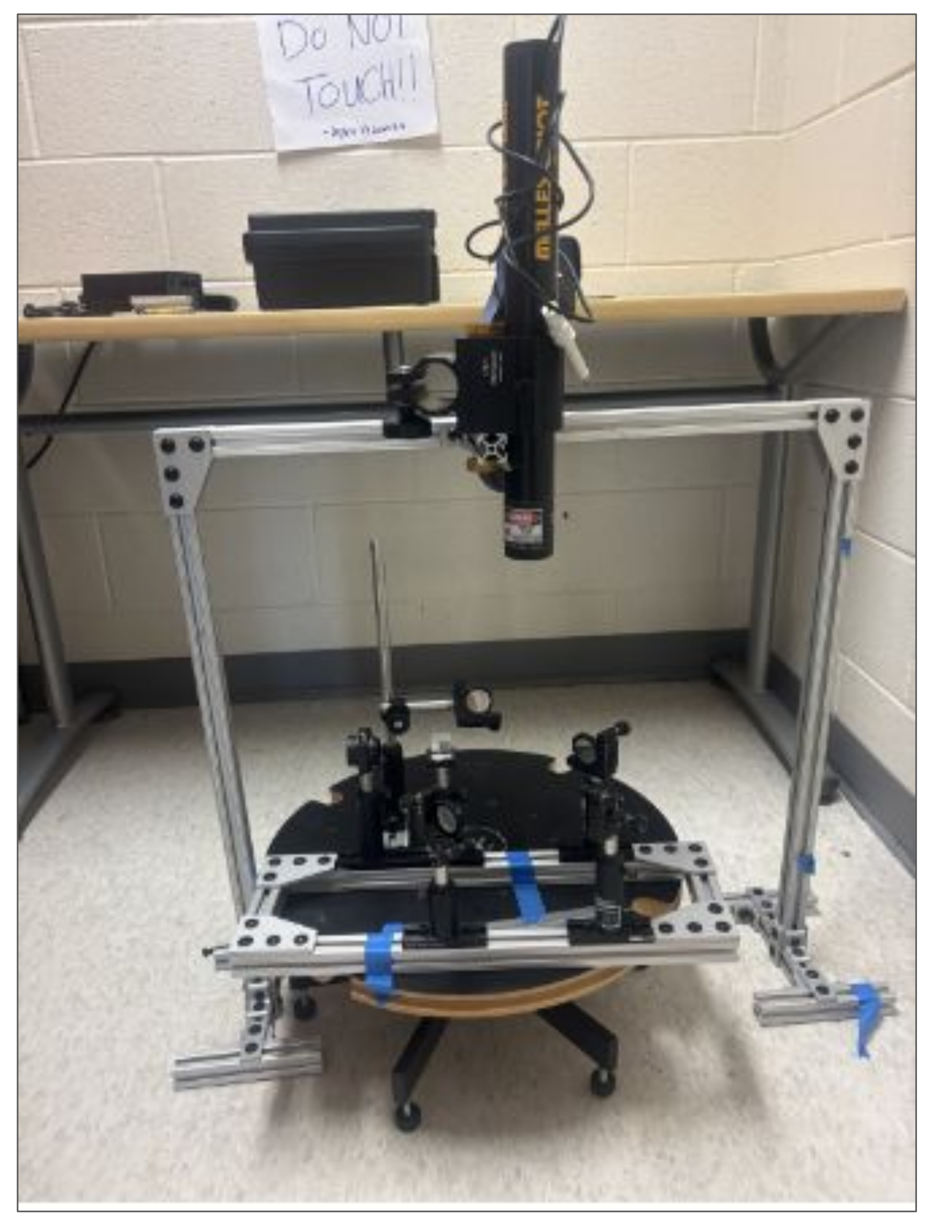
Engineering

- CAD
- Screws
- Aluminium bars
- Clamp

1. Built Sagnac prototype on Quantum Lab’s seismic isolation table
2. Planned incorporation of DC motor + suspended laser
3. CAD of SPINNI → scrapped DC motor ideas
4. Finish Sagnac prototype/testing pattern on turntable
5. Transport to Astro lab from Quantum lab
6. Ulrich
7. Center mirror problem → test detector

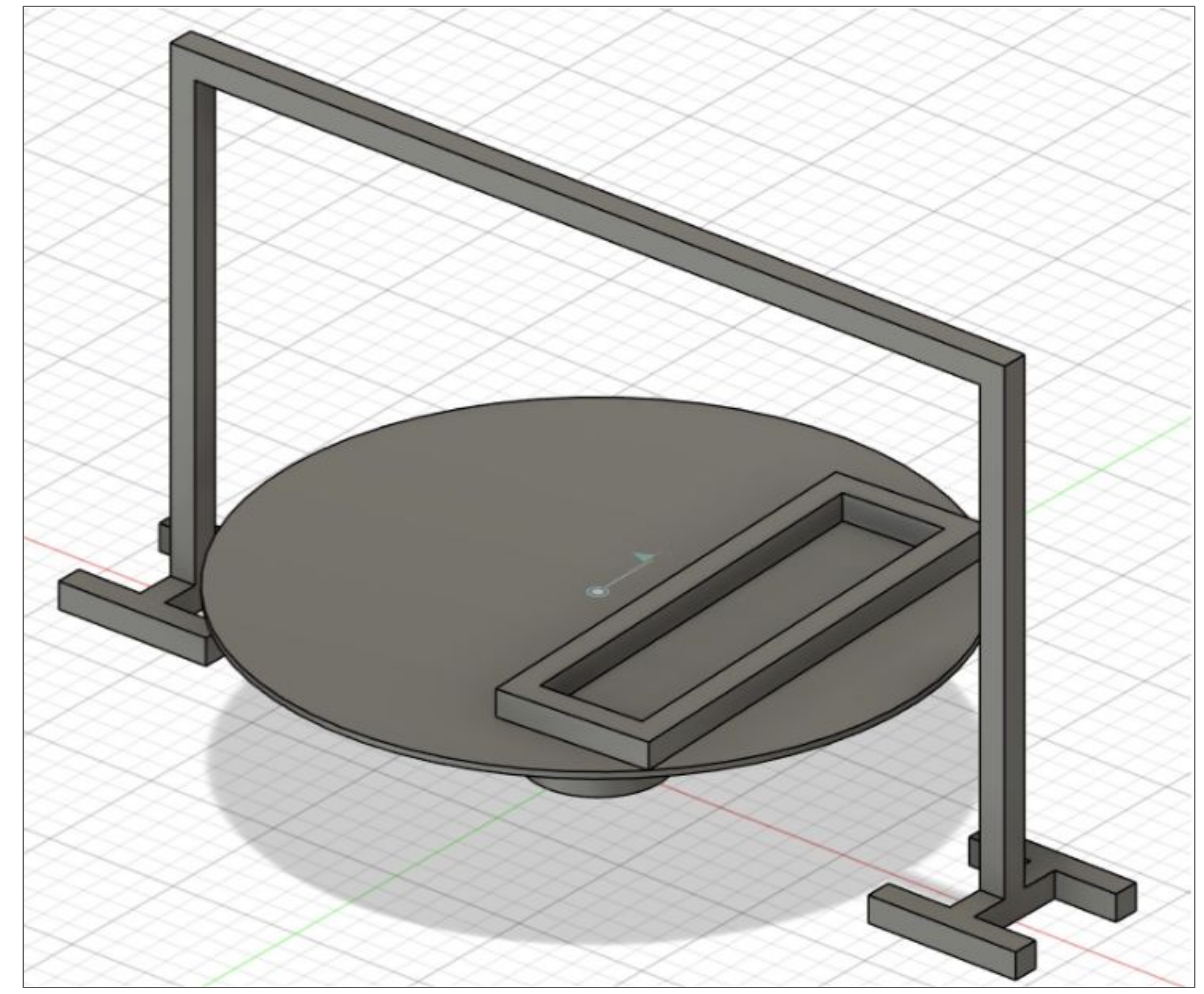
Main Findings

Increasing the portability of complex astronomical technologies through miniaturization requires high precision. **A miniaturized Sagnac interferometer is possible,** and rotational quantities of Earth are measurable, but Earth’s rotation **varies by around 10^{-8}**, making changes extremely small and **requiring extensive sensitivity** on the part of the interferometer.



Design

- Metal rig on rotating circular table
- HeNe laser chosen for long coherence length
  - Anchored above interferometer system to distribute weight
- Beam splitter divides laser into two paths
  - Angled mirrors guide beams around system
- External detector measures interference pattern
- Multi-part design easy to disassemble for future extraterrestrial applications



Discussion & Conclusions

- We were able to create a miniaturized Sagnac interferometer that effectively produces the Sagnac effect.
- Rotational rate at a smaller, controlled period…
- Smaller size makes precision extremely important
  - Require more specialized tools for our circumstances
  - Write what Ulrich said about precision here to measure the actual stuf

Acknowledgements

We would like to thank Mr. Hannum, Dr. Parilla, and Avery Conklin for their support.